

Lab 1A — AWS Console Step-by-Step Deployment Guide

This document provides exact AWS Console instructions for completing Lab 1A without using bash scripts. Follow the steps in order. Region used: us-east-1 (N. Virginia).

Step 0 – Set Region

- Open AWS Console
- Select region: N. Virginia (us-east-1)

Step 1 – Identify VPC and Subnets

- Go to VPC → Your VPCs → select lab VPC → copy VPC ID
- Go to VPC → Subnets → identify:
 - One public subnet (route to Internet Gateway)
 - Two private subnets in different AZs
- Copy subnet IDs

Step 2 – Create Security Groups

- Create EC2 Security Group (sg-ec2-lab):
 - Inbound: HTTP 80 from 0.0.0.0/0
 - Optional SSH 22 from your IP
- Create RDS Security Group (sg-rds-lab):
 - Inbound: MySQL 3306
 - Source: EC2 Security Group

Step 3 – Create RDS Subnet Group

- Go to RDS → Subnet groups → Create
- Name: lab-private-dbsubnets
- VPC: lab VPC
- Add two private subnets
- Create subnet group

Step 4 – Create RDS MySQL Instance

- Go to RDS → Databases → Create database
- Standard create → MySQL
- DB identifier: lab-mysql
- Instance class: db.t3.micro
- Public access: No
- VPC security group: sg-rds-lab
- Subnet group: lab-private-dbsubnets
- Wait until status = Available
- Copy RDS endpoint

Step 5 – Create Secrets Manager Secret

- Go to Secrets Manager → Store a new secret

- Secret type: Other type
- Keys: username, password, host, port, dbname
- Name: lab/rds/mysql
- Store secret

Step 6 – Create IAM Role for EC2

- Go to IAM → Roles → Create role
- Trusted entity: EC2
- Role name: lab-ec2-secrets-role
- Add inline policy allowing secretsmanager:GetSecretValue on lab/rds/mysql

Step 7 – Launch EC2 Instance

- Go to EC2 → Launch instance
- AMI: lab AMI
- Instance type: t3.micro
- Subnet: public subnet
- Security group: sg-ec2-lab
- IAM role: lab-ec2-secrets-role
- User data: paste 1a_user_data.sh
- Launch instance
- Copy Public IPv4 address

Step 8 – Application Test

- Open browser
- http://EC2_PUBLIC_IP/init
- http://EC2_PUBLIC_IP/add?note=first_note
- http://EC2_PUBLIC_IP/list